# 1 Submanifolds of $\mathbb{R}^n$

**Gemini's Note: Building the Intuition**
Before we tackle the formal mathematical definitions from the lecture, let's establish a geometric anchor. What exactly is a *"submanifold"*? Think of it as a smooth, lower-dimensional geometric object living inside a higher-dimensional space.

- A 1D curve drawn on a 2D sheet of paper (like a circle).
- A 2D surface floating in a 3D room (like a soap bubble or a piece of paper twisted into a cylinder).

The defining feature of any manifold is that if you zoom in close enough to any point, the surface looks *perfectly flat*. A 2D sphere looks exactly like a flat 2D plane if you are an ant standing on it. The formal mathematics below just provides us with three different ways to rigorously prove this *"locally flat"* property!

## 1.1 Submanifold Charts (The Flattening Map)

To prove a space is a manifold, we can show that we can mathematically *"flatten"* it out.

**Definition 12.10** (*Submanifold of* $\mathbb{R}^n$)**:** Given integers $0 < d < n$ and $k \geq 1$. We say that a nonempty subset $M \subset \mathbb{R}^n$ is a *"d-dimensional submanifold of* $\mathbb{R}^n$ *of class* $C^k$*"* if, for every point $p_0 \in M$, there exists:

- an open set $U \subset \mathbb{R}^n$ containing $p_0$,
- an open set $V \subset \mathbb{R}^n$ containing 0,
- and a $C^k$-diffeomorphism $\Psi : U \to V$

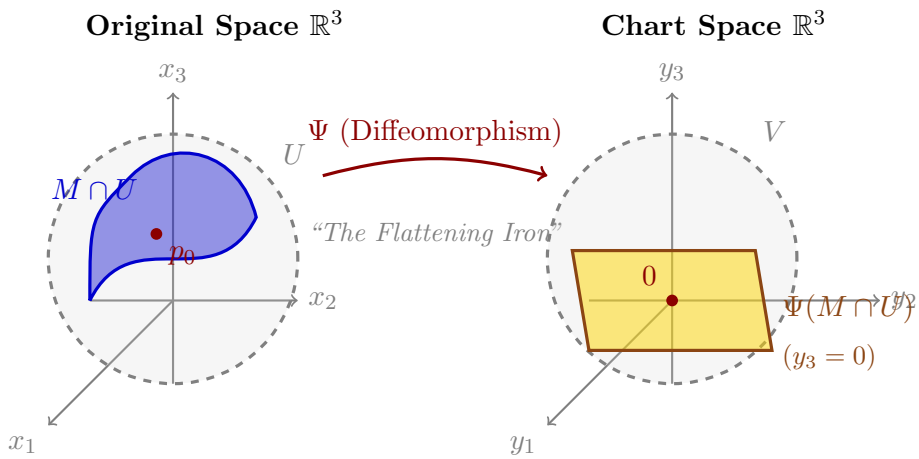
such that:

$$\Psi(M \cap U) = \{y \in V \mid y_{d+1} = y_{d+2} = \cdots = y_n = 0\}$$

The map $\Psi$ is called a **submanifold chart**.

**Gemini's Note: Unpacking the Chart**
Your TA perfectly summarizes the core concept here: *"Essentially, only* $y_1, \ldots, y_d$ *define our submanifold."*

Imagine the diffeomorphism $\Psi$ acting as a magical *"flattening iron"*. It takes the curvy, twisted piece of the manifold $M \cap U$ and perfectly irons it out flat onto the coordinate axes in the target space $V$. Because our manifold is $d$-dimensional, all the extra coordinates beyond $d$ (from $y_{d+1}$ up to $y_n$) are simply flattened to zero.



## 1.2 Equivalent Local Representations

There are multiple valid ways to mathematically describe a submanifold. Depending on the calculus problem at hand, you must flexibly switch between them.

**Proposition 12.12** (*Different Local Representations of Submanifold*)**:** Let $0 < d < n$ and $k \geq 1$ be integers, and let $M \subset \mathbb{R}^n$ be a nonempty subset. The following four are equivalent:

1. $M \subset \mathbb{R}^n$ is a $d$-dimensional submanifold.
2. **(Implicit representation):** For all $p_0 \in M$ there are $U \subset \mathbb{R}^n$ open containing $p_0$ and $f \in C^k(U, \mathbb{R}^{n-d})$ such that $Df_{p_0}$ has maximal rank (i.e., has rank $n-d$), and
$$M \cap U = \{x \in U \mid f(x) = 0\}.$$
3. **(Parametric representation):** For all $p_0 \in M$ there is $V \subset \mathbb{R}^d$ open, $y_0 \in V$ and $G \in C^k(V, \mathbb{R}^n)$ such that $G(y_0) = p_0$, $DG_{y_0}$ has maximal rank (i.e., has rank $d$), and, in addition, for all $V' \subset V$ open there is $U' \subset \mathbb{R}^n$ open such that
$$M \cap U' = G(V').$$
4. **(Graphical representation):** There are $U \subset \mathbb{R}^n$ open containing $p_0$, $V \subset \mathbb{R}^d$ open, and $g \in C^k(V, \mathbb{R}^{n-d})$ such that
$$M \cap U = P(\text{graph}(g))$$
where $P : \mathbb{R}^n \to \mathbb{R}^n$ is some permutation of the coordinates and
$$\text{graph}(g) := \{x \in V \times \mathbb{R}^{n-d} \mid (x_{d+1}, \ldots, x_n) = g(x_1, \ldots, x_d)\} \subset \mathbb{R}^n.$$

**Gemini's Note: Implicit vs. Parametric & The Power of Permutation**

Your TA's annotations cut right to the core of how to use this proposition practically:

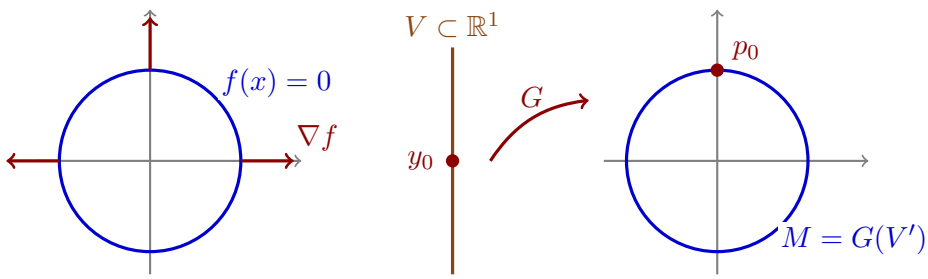
- **From (b) (Implicit):** As noted, this is *"same as **(a)** but we define the diffeo implicitly"*. You carve out the manifold by applying equations ($f(x) = 0$) to restrict the ambient space.
- **From (c) (Parametric):** The TA notes this is *"using params, most intuitive"*. You build the manifold by mapping a flat, lower-dimensional space upwards into $\mathbb{R}^n$.
- **The Power of Permutation in (d):** Any smooth curve locally looks like a function graph $y = g(x)$. But what if the curve is strictly vertical? A vertical line fails the vertical-line test and isn't a function. **This is where the permutation $P$ gives you power:** it is simply a mathematical permission slip to swap the axes! If $y = g(x)$ fails, you apply $P$ to swap the coordinates and successfully write it as $x = g(y)$ instead.

**(b) Implicit**

*Carving out by constraints*

**(c) Parametric**

*Drawing from parameters*



### 1.3 A Zoo of Submanifolds: Choosing the Right Tool

Although the four representations in Proposition 12.12 are mathematically equivalent, in practice, one method is usually vastly superior depending on the geometry of $M$.

**Example** (*The Trivial Case: A Straight Line*)**:** Consider the line $y = 2x$ in $\mathbb{R}^2$. This is the simplest possible 1D submanifold.

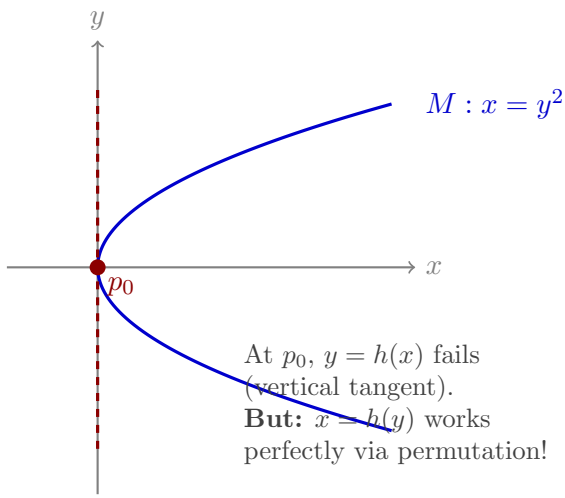
- **(Graphical):** It is already explicitly given as a global graph $g(x) = 2x$. A permutation $P$ is not even needed.
- **(Implicit):** We can define it as $f(x, y) = 2x - y = 0$. The gradient $\nabla f = (2, -1)$ trivially has rank 1 everywhere.

**Example** (*The Masterclass: The Unit Circle*)**:** The circle $M = \mathbb{S}^1 \subset \mathbb{R}^2$ $(d = 1, n = 2)$ is the perfect example because it forces us to use all tools locally:

- **(Implicit):** Defined globally by one constraint $f(x, y) = x^2 + y^2 - 1 = 0$. The Jacobian $Df = (2x, 2y)$ has maximal rank 1 everywhere on $M$.
- **(Parametric):** Easily drawn using an angle $\theta$: $G(\theta) = (\cos\theta, \sin\theta)^T$.
- **(Graphical & Permutation):** At the *"North Pole"* $(0, 1)$, the circle is locally the graph $y = g(x) = \sqrt{1 - x^2}$. However, at the *"Equator"* $(1, 0)$, the tangent is vertical. We *must* apply the permutation $P(x, y) = (y, x)$ to swap the axes, allowing us to successfully write the local graph as $x = g(y) = \sqrt{1 - y^2}$.

**Example** (*When Graphical & Permutation Win: The Sideways Parabola*)**:** Let $M = \{(x, y) \in \mathbb{R}^2 \mid x = y^2\}$.

- If we attempt to represent this classically as a **graph** $y = h(x)$ at the origin $p_0 = (0, 0)$, we fail miserably: The curve splits into $y = \pm\sqrt{x}$ and the derivative approaches infinity. The vertical tangent destroys the function property!
- **The Solution via Permutation (d):** This is where point **(d)** shines. We apply the permutation $\sigma(x, y) = (y, x)$, conceptually swapping the axes. Suddenly, the set is locally (and globally) the perfect, infinitely differentiable graph of $x = h(y) = y^2$.

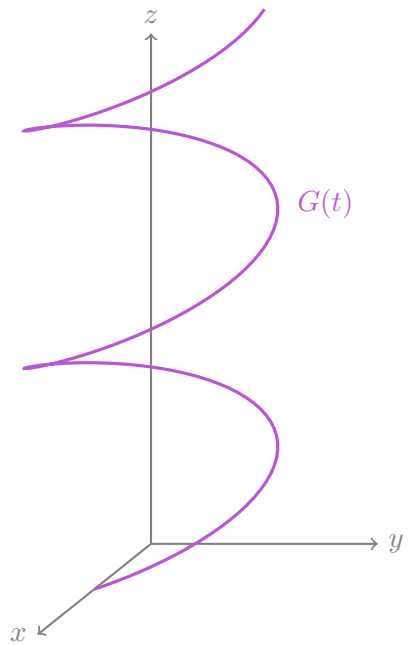



**Example** (*When Parametric Wins: The 3D Helix*)**:** Consider a spring (helix) spiraling upwards in $\mathbb{R}^3$. Attempting to represent this **implicitly** is a nightmare: You would need to define

two surfaces that perfectly intersect exactly along the spring, and prove their gradients are always linearly independent. However, the **parametric** representation **(c)** is beautifully simple. We only need $d = 1$ parameter $t$:

$$G(t) = \begin{pmatrix} \cos t \\ \sin t \\ t \end{pmatrix}$$

Its Jacobian $DG = (-\sin t, \cos t, 1)^T$ is never the zero vector (since the $z$-component is always 1), guaranteeing maximal rank 1 everywhere.




A helix is extremely easy to *"draw"* (parameterize).

**Example (***When Implicit Wins: The 2D Sphere***):** Consider the surface of a ball $M = \mathbb{S}^2 \subset \mathbb{R}^3$. Using a **parametric** representation **(c)** is problematic. If you use spherical coordinates (longitude and latitude), the math breaks down at the North and South poles (the Jacobian loses rank because all longitudes collapse to a single point). You would need multiple overlapping charts to patch the sphere together. Instead, the **implicit** representation **(b)** solves the whole surface globally with one elegant constraint ($n - d = 3 - 2 = 1$ equation):

$$f(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$$

The gradient $\nabla f = (2x, 2y, 2z)$ is never $(0, 0, 0)$ on the sphere, thus it always has full rank 1.

**Example (***The Perfect Graph: The Saddle***):** Consider the hyperbolic paraboloid (a saddle) in $\mathbb{R}^3$, defined by $z = x^2 - y^2$.

- **(Graphical):** This is an absolute dream for the graphical representation **(d)**. We need no permutation because $M$ is already globally given as an exact graph $z = g(x, y) = x^2 - y^2$.
- **(Implicit):** Also very simple: $f(x, y, z) = x^2 - y^2 - z = 0$. The gradient is $\nabla f = (2x, -2y, -1)$. Since the $z$-component is always $-1$, the vector can never be $(0, 0, 0)$. The rank is thus maximal (1) everywhere.

**Example (***The Parametric Triumph: The Torus***):** A torus (donut) is a 2D submanifold in $\mathbb{R}^3$.

- **(Implicit):** Very ugly. The equation is $(x^2 + y^2 + z^2 + R^2 - r^2)^2 - 4R^2(x^2 + y^2) = 0$. Finding the gradient and checking for zero vectors is a tedious algebraic exercise.

- **(Parametric):** Wins flawlessly! A torus is intuitively drawn using two angles (parameters). Let $\theta$ be the angle around the Z-axis and $\varphi$ be the angle around the tube:

$$G(\theta, \varphi) = \begin{pmatrix} (R + r\cos\varphi)\cos\theta \\ (R + r\cos\varphi)\sin\theta \\ r\sin\varphi \end{pmatrix}$$

  The dimension of the parameter space is $d = 2$, which perfectly matches the dimension of the manifold.

**Example** (*The Anti-Example: The Double Cone*)**:** To understand why the *"maximal rank"* condition in Proposition 12.12 is so crucial, consider the double cone:

$$M = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - z^2 = 0\}$$

Is this a 2D submanifold? Let's check condition **(b)** (Implicit): We have $f(x, y, z) = x^2 + y^2 - z^2 = 0$. We compute the gradient:

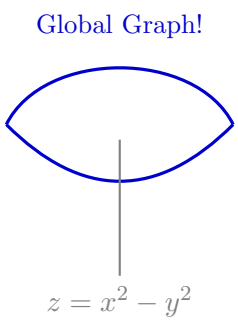
$$\nabla f = (2x, 2y, -2z)$$

Now consider the origin $p_0 = (0, 0, 0)$, which clearly lies on the cone. There we get:

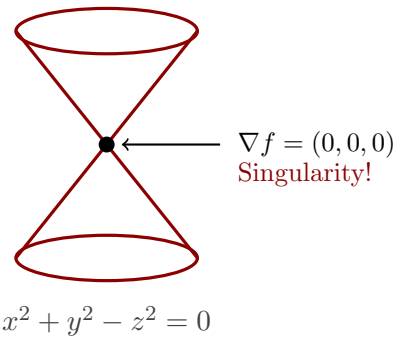
$$\nabla f(0, 0, 0) = (0, 0, 0)$$

The rank of the Jacobian matrix (here the gradient) drops to 0 at the origin. It is not maximal! **Conclusion:** The cone is *not* a manifold at the origin. Geometrically, this makes sense: At the origin, the cone has an infinitely sharp singularity. It cannot be *"ironed flat"* as required by condition **(a)**.

**The Saddle (Graphical)**




**The Cone (Anti-Example)**



# 2 Tangent Spaces

**Teacher's Note:** Before we dive into the heavy formulas, let's build some intuition. Imagine you are walking on a smooth, curved mountain (a manifold). At any exact point you stand, if you look straight ahead, left, or right, you are looking along a flat plane that perfectly grazes the mountain at your feet. That flat plane is the *"tangent space"*. The direction sticking straight up into the sky, perpendicular to your flat plane, is the *"normal"* vector.

## 2.1 Formal Definition

**Definition 12.13 (*Tangent and Normal Vectors*):** Let $M \subset \mathbb{R}^n$ be a $d$-dimensional $C^1$ submanifold. For a given point $p \in M$, we say that a vector $\tau \in \mathbb{R}^n$ is *"tangent"* to $M$ at $p$ if there exists a sequence $p_k \in M$ with $p_k \to p$, $p_k \neq p$, and a sequence of positive numbers $r_k$ converging to zero such that

$$\lim_{k\to\infty} \frac{p_k - p}{r_k} = \tau.$$

The set of all tangent vectors to $M$ at $p$ is called the *"tangent space"* of $M$ at $p$ and is denoted by

$$T_pM := \{\tau \in \mathbb{R}^n \mid \tau \text{ is } \textit{"tangent"} \text{ to } M \text{ at } p\}.$$

Furthermore, we say that a vector $\nu \in \mathbb{R}^n$ is *"normal"* (i.e., perpendicular) to $M$ at $p$ if

$$\tau \cdot \nu = 0 \quad \text{for all } \tau \in T_pM.$$

The set of all normal vectors at $p$ is called the *"normal space"* of $M$ at $p$.

This is the formal definition of *"tangent"* and *"normal"* spaces at a point $p$ of some given $C^1$ submanifold. However, as you saw in the lectures, there is a much better way to find the *"tangent space"* using partial derivatives.

## 2.2 Practical Computation for Parametrized Surfaces

For parametrized surfaces (essentially 2-dimensional submanifolds in $\mathbb{R}^n$), the best way to compute the *"tangent space"* and *"normal space"* at any point is using the following proposition.

**Proposition 12.16 (*Tangent and Normal Vectors for Parametrized Surfaces*):** Let $V \subset \mathbb{R}^2$ be open and let $\phi : V \to \mathbb{R}^3$ be a parametrized surface. Then, for every $x \in V$ and $p = \phi(x) \in M$, the following hold:

1. The *"tangent space"* to $M$ at $p$ is given by the span of the partial derivatives:

$$T_pM = D\phi_x(\mathbb{R}^2) = \operatorname{span}\{\partial_1\phi(x), \partial_2\phi(x)\}.$$

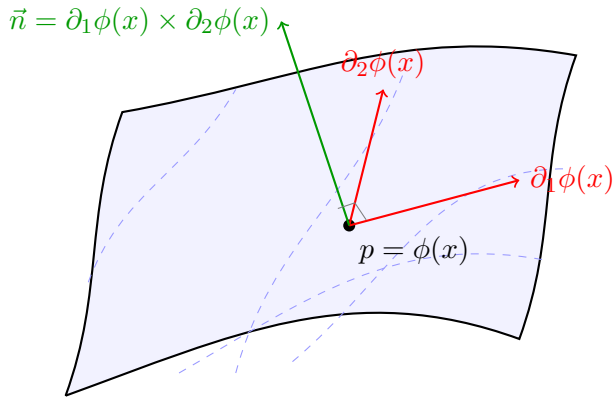
2. The vectors $\partial_1\phi(x)$ and $\partial_2\phi(x)$ are linearly independent and form a basis of $T_pM$.

3. A *"normal vector"* to $M$ at $p = \phi(x)$ is given by the cross product $\partial_1\phi(x) \times \partial_2\phi(x)$, and a *"normal vector"* with modulus one (unit normal) is given by

$$\nu(x) := \frac{\partial_1\phi(x) \times \partial_2\phi(x)}{|\partial_1\phi(x) \times \partial_2\phi(x)|}.$$

4. Moreover, the *"normal space"* is the span of this vector:

$$N_pM = \operatorname{span}\{\nu(x)\}.$$




*Visualization of Proposition 12.16*

### 2.3 Examples

Let's apply this to a foundational example from the notes.

**Example (*Spherical Coordinates*):** Consider the parameterization of a unit sphere with a thin slice deleted (the boundary of the unit ball):

$$\phi : (0, \pi) \times (-\pi, \pi) \to \mathbb{R}^3, \quad (\theta, \varphi) \mapsto \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

Let $p = \phi(x) \in M$ for some $x = (\theta, \varphi) \in (0, \pi) \times (-\pi, \pi)$. The *"tangent space"* is $T_pM = D\phi_x(\mathbb{R}^2)$. We compute the partial derivatives spanning $T_pM$:

$$\partial_1\phi(x) = \frac{\partial}{\partial\theta}\begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix},$$

$$\partial_2\phi(x) = \frac{\partial}{\partial\varphi}\begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}.$$

A *"normal vector"* is then given by the cross product $\vec{n} = \partial_1\phi(x) \times \partial_2\phi(x)$:

$$\begin{aligned} \vec{n} &= \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix} \times \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix} \\ &= \begin{pmatrix} (\cos\theta\sin\varphi)(0) - (-\sin\theta)(\sin\theta\cos\varphi) \\ (-\sin\theta)(-\sin\theta\sin\varphi) - (\cos\theta\cos\varphi)(0) \\ (\cos\theta\cos\varphi)(\sin\theta\cos\varphi) - (\cos\theta\sin\varphi)(-\sin\theta\sin\varphi) \end{pmatrix} \\ &= \begin{pmatrix} \sin^2\theta\cos\varphi \\ \sin^2\theta\sin\varphi \\ \cos\theta\sin\theta\cos^2\varphi + \cos\theta\sin\theta\sin^2\varphi \end{pmatrix}. \end{aligned}$$

**Teacher's Note:** Notice how in the z-component, we can factor out $\cos\theta\sin\theta$, leaving $(\cos^2\varphi + \sin^2\varphi)$, which simplifies to 1.

Thus, the *"normal vector"* simplifies beautifully to:

$$\vec{n} = \begin{pmatrix} \sin^2\theta\cos\varphi \\ \sin^2\theta\sin\varphi \\ \sin\theta\cos\theta \end{pmatrix} = \sin\theta\begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

The magnitude of this vector is $|\vec{n}| = \sin\theta$. Therefore, the unit *"normal vector"* $\nu(x)$ is simply the position vector itself:

$$\nu(x) := \frac{\vec{n}}{|\vec{n}|} = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix} = \phi(\theta,\varphi) = \phi(x).$$

Geometrically, this makes perfect sense! For a sphere centered at the origin, the direction pointing strictly outward from the surface is exactly the same direction as the position vector of the point itself.

**Example (*Teacher's Extra Practice: The Paraboloid*):** To make sure you really have the hang of this, let's look at another classic surface: the paraboloid given by $z = x^2 + y^2$. We can parameterize this surface by simply using $x$ and $y$ as our parameters. Let $\phi(x,y) = (x, y, x^2 + y^2)^T$.

1. **Find the Tangent Vectors:**

$$\partial_x\phi(x,y) = \begin{pmatrix} 1 \\ 0 \\ 2x \end{pmatrix}, \quad \partial_y\phi(x,y) = \begin{pmatrix} 0 \\ 1 \\ 2y \end{pmatrix}$$

These two vectors form the basis of the *"tangent space"* $T_pM$.

2. **Find the Normal Vector:**

$$\vec{n} = \partial_x\phi \times \partial_y\phi = \begin{pmatrix} 1 \\ 0 \\ 2x \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 2y \end{pmatrix} = \begin{pmatrix} -2x \\ -2y \\ 1 \end{pmatrix}$$

This vector spans the *"normal space"* $N_pM$ at any point on the paraboloid!

**Example (*Teacher's Extra Practice: The Cylinder*):** Let's consider a cylinder of radius $R$, described by $x^2 + y^2 = R^2$. A natural parameterization uses the angle $\theta$ and the height $z$:

$$\phi(\theta, z) = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ z \end{pmatrix}.$$

1. **Find the Tangent Vectors:** Taking partial derivatives with respect to our parameters $\theta$ and $z$:

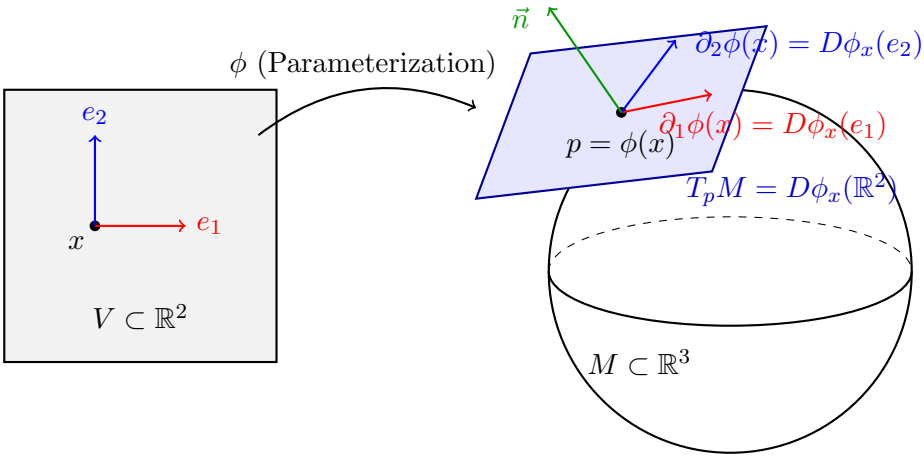
$$\partial_\theta\phi(\theta,z) = \begin{pmatrix} -R\sin\theta \\ R\cos\theta \\ 0 \end{pmatrix}, \quad \partial_z\phi(\theta,z) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Notice that $\partial_z\phi$ points straight up (parallel to the z-axis), and $\partial_\theta\phi$ is *"tangent"* to the circular cross-section.

2. **Find the Normal Vector:** Taking the cross product:

$$\vec{n} = \partial_\theta\phi \times \partial_z\phi = \begin{pmatrix} -R\sin\theta \\ R\cos\theta \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} R\cos\theta \\ R\sin\theta \\ 0 \end{pmatrix}$$

The *"normal vector"* points directly away from the z-axis in the xy-plane, completely ignoring the z-height. Its magnitude is $|\vec{n}| = \sqrt{R^2\cos^2\theta + R^2\sin^2\theta} = R$. Thus, the unit normal is simply $(\cos\theta, \sin\theta, 0)^T$.




*Visualization: Pushing forward the basis vectors to span the tangent space.*

# 3 Jordan Measure

**Teacher's Note: The *"Pixel"* Approach to Area**
Before we look at the formal math, let's think about computer screens. How does a computer display a circle? It uses pixels! If the pixels are huge, the circle looks like a blocky mess. If the pixels are tiny, it looks perfectly smooth.

Jordan measure is exactly this idea. We want to find the volume of strange, curvy shapes by approximating them with blocky *"pixels"* (which we call *"dyadic cubes"*).

- If we take all the pixels that touch the shape (even a little bit), we get an *"outer approximation"* (an overestimate).
- If we only take the pixels that are 100% inside the shape, we get an *"inner approximation"* (an underestimate).

If we make our pixels infinitely small, and the inner and outer approximations squeeze together to the exact same number, we say the shape is *"Jordan measurable"*!

## 3.1 Dyadic Cubes: Intuition vs. Script

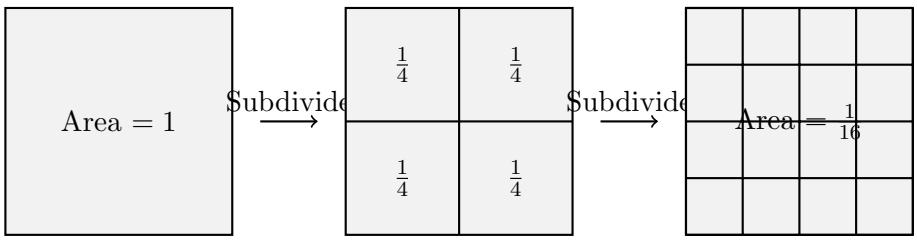
To build our grid, we use a very specific type of building block. Here is how your TA intuitively described it:

**Definition (*TA's Intuitive Dyadic Cubes*):** A *"dyadic cube"* of side length $2^{-p}$ (for some integer $p \geq 0$) is defined on a standard coordinate grid. The union of finite, same-size dyadic cubes is called a *"dyadic set"*.

For a fixed *"pixel size"* $2^{-p}$, a dyadic set $E$ in $\mathbb{R}^n$ can be represented mathematically as:

$$E = \bigcup_{l=1}^{N} 2^{-p} \left( a(l) + [0,1)^n \right),$$

where $a(l) \in \mathbb{Z}^n$ is an integer vector representing the coordinates of the bottom-left corner of each cube in the grid.




$p = 0$
*This is one dyadic cube of side length* $1 = 2^0$

$p = 1$
*Side length* $1/2 = 2^{-1}$

$p = 2$
*Side length* $1/4 = 2^{-2}$

Any dyadic cube can be perfectly subdivided into smaller cubes. If we want to compare two different dyadic sets that have different pixel sizes, we can always choose the smallest pixel size and subdivide the larger cubes to match.

Now, let's look at the rigorous definition from the official script:

**Definition 13.1 (*Dyadic Cubes*):** A *"dyadic cube"* of side length $2^{-p}$ ($p \in \mathbb{N}_0$) in $\mathbb{R}^n$ is a set of the form

$$Q = 2^{-p}(a + [0,1)^n),$$

where $a \in \mathbb{Z}^n$. We denote the set of all dyadic cubes of side length $2^{-p}$ by $\Omega_p$.

**Why half-open intervals $[0,1)^n$?**
You might wonder why the definition uses a bracket and a parenthesis. If we used closed intervals $[0,1] \times [0,1]$ for our cubes, their boundaries would overlap. For example, the right edge of the left cube would be the exact same line as the left edge of the right cube.

By defining them as $[0,1)$, we mean *"include the left and bottom edges, but leave the top and right edges open"*. This ensures that every single point in $\mathbb{R}^n$ belongs to exactly *one* unique dyadic cube. No double counting!

## 13.2 Inner and Outer Volumes

Similar to the Riemann Integral in 1D, we define the volume (measure) of an arbitrary set $A \subset \mathbb{R}^n$ by approximating it from the outside and from the inside. First, note that the volume (or measure $\mu$) of a single dyadic cube in $\mathbb{R}^n$ at level $p$ is simply $\mu(Q) = 2^{-pn}$.

Here is the rigorous script definition for these approximations:

**Definition 13.2 (*Inner and Outer Approximations*):** Let $A \subset \mathbb{R}^n$ be a bounded set. For $p \in \mathbb{N}_0$, we define the **inner approximation** $I(A,p)$ and the **outer approximation** $O(A,p)$ by:

$$I(A,p) := \bigcup \{Q \in \Omega_p \mid \overline{Q} \subset A^\circ\},$$
$$O(A,p) := \bigcup \{Q \in \Omega_p \mid Q \cap A \neq \emptyset\}.$$

Here, $\overline{Q}$ is the closure of the cube, and $A^\circ$ is the interior of $A$.

**Definition 13.3 (*Jordan Measure*):** The **inner measure** and **outer measure** of $A$ are defined as the limits of the volumes of these approximations:

$$\mu_{\text{in}}(A) := \lim_{p\to\infty} \mu(I(A,p)), \quad \text{and} \quad \mu_{\text{out}}(A) := \lim_{p\to\infty} \mu(O(A,p)).$$

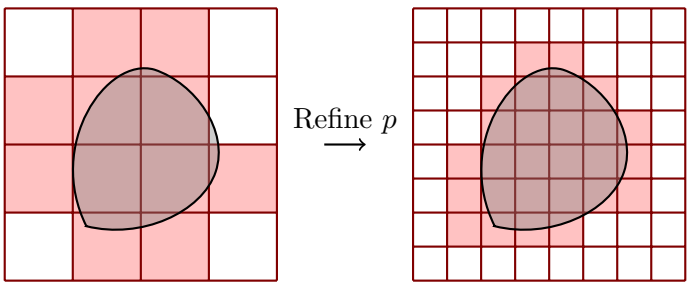
A bounded set $A \subset \mathbb{R}^n$ is called **Jordan measurable** if $\mu_{\text{in}}(A) = \mu_{\text{out}}(A)$. In this case, this common value is called the Jordan measure of $A$, denoted by $\mu(A)$.

**Teacher's Note: Bridging the Script to the TA Notes**
Wow, the script's definition looks scary! But it is exactly what we drew in the TA notes (where the TA used $F_i$ for inner and $G_i$ for outer approximations).

- The script's $p$ (level of subdivision) is just the TA's index $i$.
- The script's $O(A,p)$ is the TA's $G_i$. The condition $Q \cap A \neq \emptyset$ literally just means *"include the cube if it touches the set $A$ at all."*
- The script's $I(A,p)$ is the TA's $F_i$. The condition $\overline{Q} \subset A^\circ$ is just a rigorous topological way of saying *"the cube must be 100% safely inside the boundaries of $A$."*

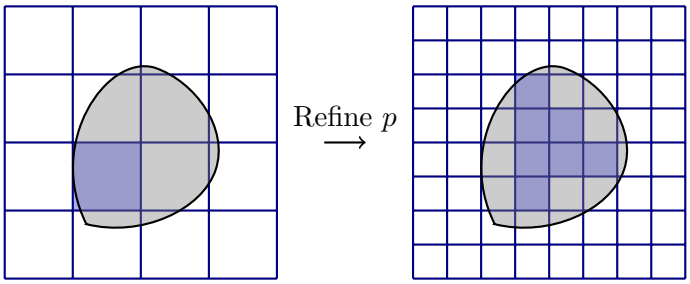
So whenever you see $I(A,p)$ and $O(A,p)$ on the exam, just picture the blue and red pixel grids below!

**Outer Approx $O(A,1)$** **Outer Approx $O(A,2)$**




Large overestimate. $\mu(O(A,2)) \leq \mu(O(A,1))$

**Inner Approx** $I(A,1)$ **Inner Approx** $I(A,2)$




Small underestimate. $\mu(I(A,2)) \geq \mu(I(A,1))$

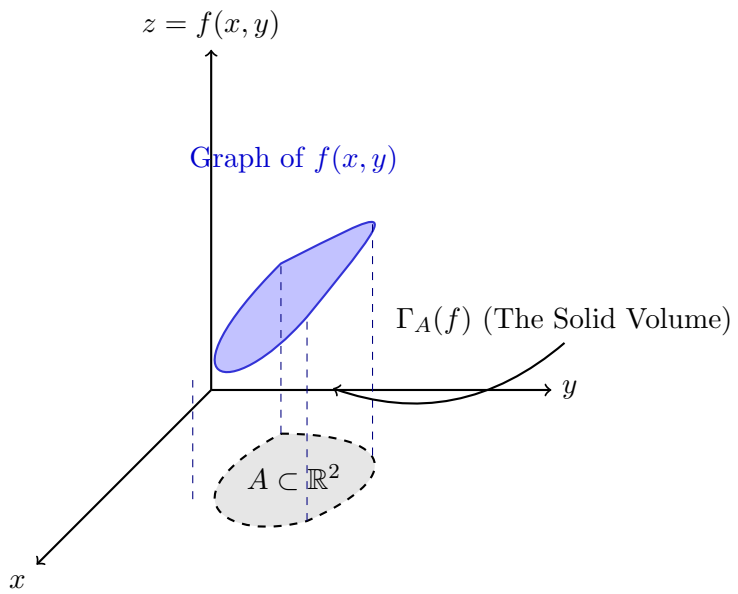
# 4 Multidimensional Integrals

One powerful way to understand the Riemann integral of a function $f : A \to \mathbb{R}$ in higher dimensions is to view it as calculating the volume *under* the graph of the function. The script formalizes this using the concept of a *"hypograph"*, which your TA sketched in the notes as the volume trapped above a domain $A$.

**Definition 13.40 (*Hypograph*):** Let $A \subset \mathbb{R}^n$ and $f : A \to [0, \infty)$. The **hypograph** of $f$ is defined as the set

$$\Gamma_A(f) := \{(x, y) \in \mathbb{R}^n \times \mathbb{R} \mid x \in A, 0 \leq y < f(x)\} \subset \mathbb{R}^{n+1}.$$

**Gemini 3.1's Note: What is a Hypograph?**
Don't let the Greek prefix intimidate you! *"Hypo"* just means *"under"* (like hypothermia means under-heating). In Calc 1, when you integrated $f(x)$ from $a$ to $b$, you were finding the 2D area *under* the curve. The hypograph is literally just that geometric shape: the region trapped between the floor ($z = 0$) and the ceiling ($z = f(x, y)$).




**Lemma 13.42 (*Riemann Integral as Jordan Measure*):** Let $A \subset \mathbb{R}^n$ be a set. A function $f : A \to [0, \infty)$ is Riemann integrable if and only if the hypograph $\Gamma_A(f)$ is Jordan measurable in $\mathbb{R}^{n+1}$. In such a case,

$$\int_A f = \mu_{n+1}(\Gamma_A(f)).$$

**Corollary 13.43 (*Integrability of Continuous Functions*):** Suppose that $E \subset \mathbb{R}^n$ is a Jordan measurable set and $f : \overline{E} \to [0, \infty)$ is a continuous function. Then, $f$ is Riemann integrable over $E$.

**TA's Survival Rule:** Thanks to Corollary 13.43, in practice, nonnegative continuous functions are almost always integrable on nicely behaved (Jordan measurable) closed sets!

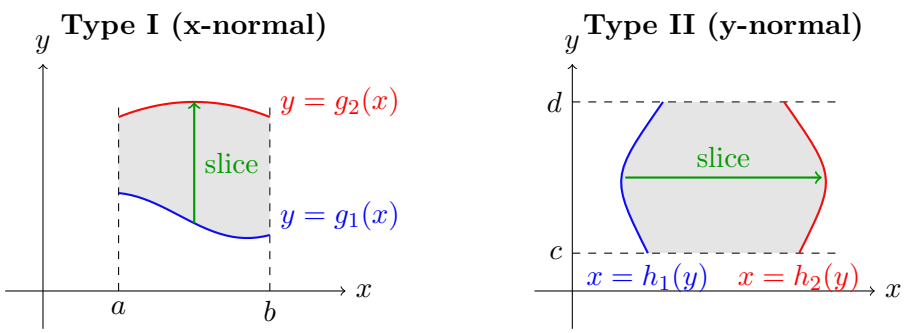
## 4.1 Normal Domains and Fubini's Theorem

Defining the integral via the hypograph is mathematically rigorous, but it doesn't help us actually calculate a number. To compute multidimensional integrals, we need to slice the domain into standard 1D integrals. We do this using **Normal Domains**.

**Definition** (*Normal Domains / Projectable Regions*)**:** There are two primary ways to slice a 2D region $A \subset \mathbb{R}^2$:

1. **Type I (x-normal):** The domain is bounded left/right by constants $a, b$, and top/bottom by functions of $x$:
$$A = \{(x, y) \in \mathbb{R}^2 \mid a \leq x \leq b, \text{ and } g_1(x) \leq y \leq g_2(x)\}$$

2. **Type II (y-normal):** The domain is bounded bottom/top by constants $c, d$, and left/right by functions of $y$:
$$A = \{(x, y) \in \mathbb{R}^2 \mid c \leq y \leq d, \text{ and } h_1(y) \leq x \leq h_2(y)\}$$



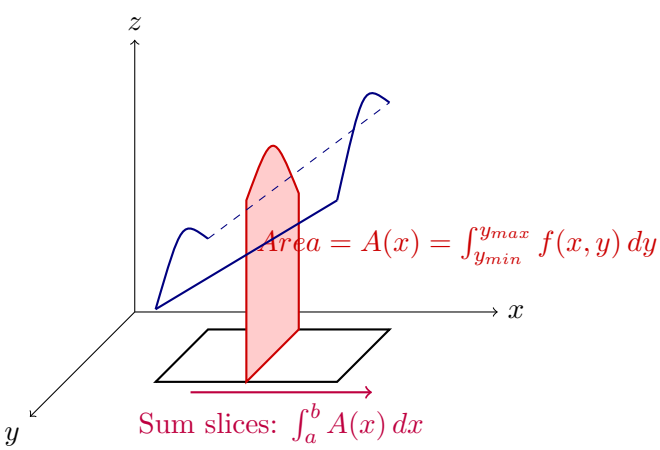

**Theorem** (*Fubini's Theorem for Normal Domains*)**:** Let $A \subset \mathbb{R}^2$ be a normal domain with respect to the x-axis (Type I). If $f : A \to \mathbb{R}$ is continuous, the double integral can be evaluated as an iterated integral:
$$\iint_A f(x, y)\, dA = \int_a^b \left[ \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy \right] dx.$$
If $A$ is Type II, we switch the order: integrate $dx$ first, then $dy$ on the outside.

**Gemini 3.1's Note: The *"Loaf of Bread"* Method**

Fubini's theorem is the absolute backbone of Calculus 2 computations. If the integral is asking for the volume of a loaf of bread, Fubini says:

1. First, pick a specific spot along the crust ($x$).
2. Slice the bread vertically there, and find the area of that one 2D slice (the inner integral $\int dy$).
3. Add up all the 2D slices from the front of the loaf to the back (the outer integral $\int dx$).

**Golden Rule of Fubini:** The outermost integral bounds **must always be constants!** If you end up with variables in your final evaluation step, you've mixed up your order!



## 4.2 Gemini 3.1's Extra Practice Examples

To truly master this, we need to do the math. Here are 4 classic exams-style examples showing exactly how to set up and execute these integrals.

**Example (*Type I Normal Domain: The Triangle*):** Evaluate $\iint_A 2xy\,dA$, where $A$ is the triangle bounded by the x-axis ($y = 0$), the line $y = x$, and the vertical line $x = 1$.

**Step 1: Understand the Domain.** This is a perfect Type I domain. The x-values go from $a = 0$ to $b = 1$. For any fixed $x$, the y-values go from the floor ($y = 0$) up to the ceiling ($y = x$).

**Step 2: Set up Fubini and Integrate.**

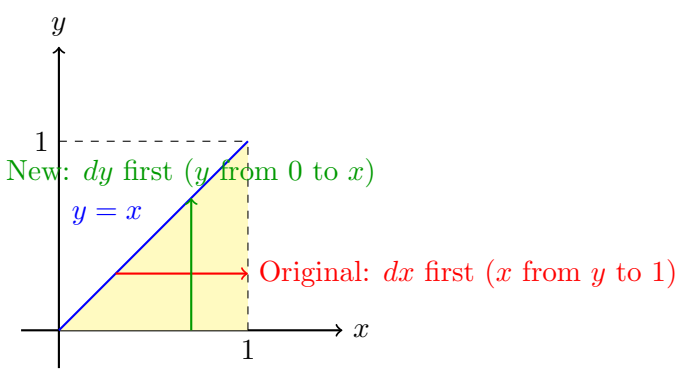
$$\begin{aligned}
\iint_A 2xy\,dA &= \int_0^1 \left( \int_0^x 2xy\,dy \right) dx \\
&= \int_0^1 \left[ xy^2 \right]_{y=0}^{y=x} dx \\
&= \int_0^1 \left( x(x)^2 - x(0)^2 \right) dx \\
&= \int_0^1 x^3\,dx \\
&= \left[ \frac{1}{4}x^4 \right]_0^1 = \frac{1}{4}.
\end{aligned}$$

**Example (*The Classic "Switching Bounds" Trap*):** Evaluate the integral:

$$I = \int_0^1 \int_y^1 e^{x^2}\,dx\,dy$$

**The Problem:** If you try to integrate $e^{x^2}$ with respect to $x$, you will get stuck immediately. It has no elementary antiderivative!

**The Solution:** We must switch the order of integration. But we **cannot** just blindly swap the bounds to $\int_y^1 \int_0^1$. We have to redraw the domain.



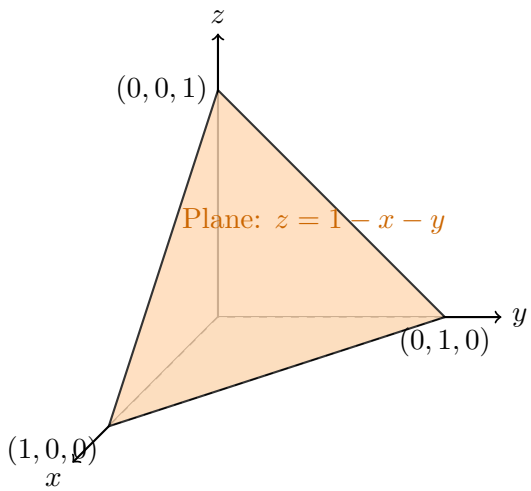

By looking at the diagram, if we want to integrate with respect to $y$ first (vertical slices), the y-bounds go from the floor ($y = 0$) to the slanted roof ($y = x$). The outer x-bounds are constants from 0 to 1.

$$
\begin{aligned}
I &= \int_0^1 \left( \int_0^x e^{x^2}\, dy \right) dx \\
&= \int_0^1 \left[ y e^{x^2} \right]_{y=0}^{y=x} dx \\
&= \int_0^1 x e^{x^2}\, dx
\end{aligned}
$$

Now, this is an easy Calc 1 substitution! Let $u = x^2$, so $du = 2x\, dx$, meaning $x\, dx = \frac{1}{2} du$.

$$
I = \int_0^1 \frac{1}{2} e^u\, du = \frac{1}{2}(e^1 - e^0) = \frac{1}{2}(e - 1).
$$

**Example (*Triple Integrals and 3D Volumes*):** Find the volume of the solid tetrahedron bounded by the coordinate planes ($x = 0, y = 0, z = 0$) and the plane $x + y + z = 1$.




**Step 1: Set the bounds.**

- **Inner (z):** Travels from the floor ($z = 0$) to the slanted roof ($z = 1 - x - y$).
- **Middle (y):** Look down at the shadow on the xy-plane. It's a triangle bounded by $y = 0$ and the line $x + y = 1 \implies y = 1 - x$.
- **Outer (x):** The absolute constants on the x-axis are from $x = 0$ to $x = 1$.

**Step 2: Integrate.**

$$
\begin{aligned}
\text{Volume} &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} 1\, dz\, dy\, dx \\
&= \int_0^1 \int_0^{1-x} (1 - x - y)\, dy\, dx \\
&= \int_0^1 \left[ (1-x)y - \frac{1}{2}y^2 \right]_0^{1-x} dx \\
&= \int_0^1 \left( (1-x)(1-x) - \frac{1}{2}(1-x)^2 \right) dx \\
&= \int_0^1 \frac{1}{2}(1-x)^2\, dx \\
&= \left[ -\frac{1}{6}(1-x)^3 \right]_0^1 = -\frac{1}{6}(0) - \left( -\frac{1}{6}(1) \right) = \frac{1}{6}.
\end{aligned}
$$

# 5 Tangent Spaces

**Teacher's Note:** Before we dive into the heavy formulas, let's build some intuition. Imagine you are walking on a smooth, curved mountain (a manifold). At any exact point you stand, if you look straight ahead, left, or right, you are looking along a flat plane that perfectly grazes the mountain at your feet. That flat plane is the *"tangent space"*. The direction sticking straight up into the sky, perpendicular to your flat plane, is the *"normal"* vector.

## 5.1 Formal Definition

**Theorem 12.13 (***Tangent and Normal Vectors***):** Let $M \subset \mathbb{R}^n$ be a $d$-dimensional $C^1$ submanifold. For a given point $p \in M$, we say that a vector $\tau \in \mathbb{R}^n$ is *"tangent"* to $M$ at $p$ if there exists a sequence $p_k \in M$ with $p_k \to p$, $p_k \neq p$, and a sequence of positive numbers $r_k$ converging to zero such that

$$\lim_{k\to\infty} \frac{p_k - p}{r_k} = \tau.$$

The set of all tangent vectors to $M$ at $p$ is called the *"tangent space"* of $M$ at $p$ and is denoted by

$$T_pM := \{\tau \in \mathbb{R}^n \mid \tau \text{ is } \textit{"tangent"} \text{ to } M \text{ at } p\}.$$

Furthermore, we say that a vector $\nu \in \mathbb{R}^n$ is *"normal"* (i.e., perpendicular) to $M$ at $p$ if

$$\tau \cdot \nu = 0 \quad \text{for all } \tau \in T_pM.$$

The set of all normal vectors at $p$ is called the *"normal space"* of $M$ at $p$.

This is the formal definition of *"tangent"* and *"normal"* spaces at a point $p$ of some given $C^1$ submanifold. However, as you saw in the lectures, there is a much better way to find the *"tangent space"* using partial derivatives.

## 5.2 Practical Computation for Parametrized Surfaces

**Theorem 12.16 (***Tangent and Normal Vectors for Parametrized Surfaces***):** Let $V \subset \mathbb{R}^2$ be open and let $\phi : V \to \mathbb{R}^3$ be a parametrized surface. Then, for every $x \in V$ and $p = \phi(x) \in M$, the following hold:

1. The *"tangent space"* to $M$ at $p$ is given by the span of the partial derivatives:

$$T_pM = D\phi_x(\mathbb{R}^2) = \text{span}\{\partial_1\phi(x), \partial_2\phi(x)\}.$$

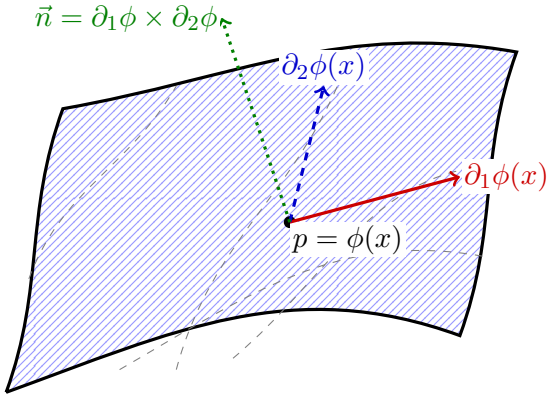
2. The vectors $\partial_1\phi(x)$ and $\partial_2\phi(x)$ are linearly independent and form a basis of $T_pM$.

3. A *"normal vector"* to $M$ at $p = \phi(x)$ is given by the cross product $\partial_1\phi(x) \times \partial_2\phi(x)$, and a *"normal vector"* with modulus one (unit normal) is given by

$$\nu(x) := \frac{\partial_1\phi(x) \times \partial_2\phi(x)}{|\partial_1\phi(x) \times \partial_2\phi(x)|}.$$

4. Moreover, the *"normal space"* is the span of this vector:

$$N_pM = \text{span}\{\nu(x)\}.$$




*Visualization of Proposition 12.16*

**Example (*Spherical Coordinates*):** Consider the parameterization of a unit sphere with a thin slice deleted:

$$\phi : (0, \pi) \times (-\pi, \pi) \to \mathbb{R}^3, \quad (\theta, \varphi) \mapsto \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

The *"tangent space"* is $T_pM = D\phi_x(\mathbb{R}^2)$. We compute the partial derivatives spanning $T_pM$:

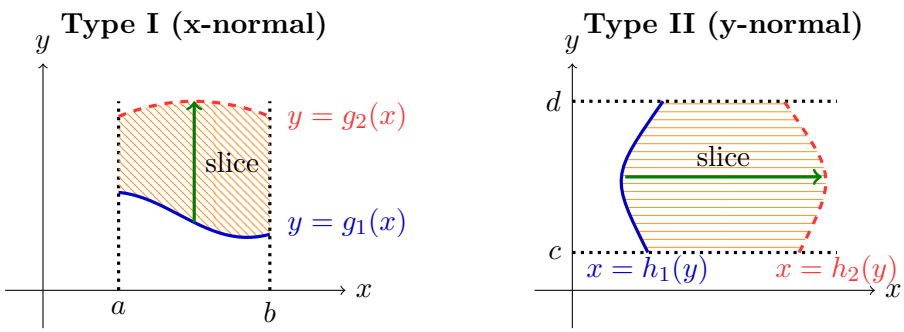
$$\partial_1\phi(x) = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix}, \quad \partial_2\phi(x) = \begin{pmatrix} -\sin\theta\sin\varphi \\ \sin\theta\cos\varphi \\ 0 \end{pmatrix}.$$

A *"normal vector"* is then given by the cross product $\vec{n} = \partial_1\phi(x) \times \partial_2\phi(x)$:

$$\vec{n} = \begin{pmatrix} \sin^2\theta\cos\varphi \\ \sin^2\theta\sin\varphi \\ \cos\theta\sin\theta\cos^2\varphi + \cos\theta\sin\theta\sin^2\varphi \end{pmatrix} = \sin\theta \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}.$$

Therefore, the unit *"normal vector"* $\nu(x)$ is simply the position vector itself $\nu(x) = \phi(\theta, \varphi)$.

**Alternative Notations & Wikipedia Warning:**
Depending on the textbook, you might see the tangent space denoted as $T_pM$, $T_xM$, or $TM_p$. The differential (Jacobian map) $D\phi_x$ is also frequently written as $d\phi_x$ or $\phi_{*,x}$ (the *"pushforward"* map). When the notes say $T_pM = D\phi_x(\mathbb{R}^2)$, they mean that the tangent plane is the image of the flat 2D plane under the linear transformation given by the Jacobian matrix.




## 6 Jordan Measure

**Teacher's Note: The *"Pixel"* Approach to Area**
We want to find the volume of strange shapes by approximating them with blocky *"pixels"* (dyadic cubes).

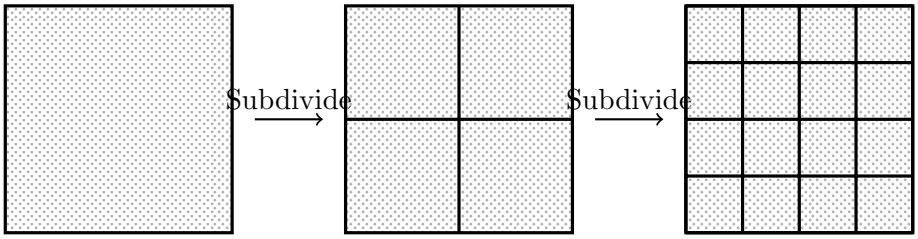
- If we take all pixels touching the shape, we get an *"outer"* overestimate.
- If we only take pixels 100% inside the shape, we get an *"inner"* underestimate.

If we make our pixels infinitely small, and the inner and outer approximations squeeze to the exact same number, the shape is *"Jordan measurable"*!

## 6.1 Dyadic Cubes: Intuition vs. Script

**Theorem (***TA's Intuitive Dyadic Cubes***):** A *"dyadic cube"* of side length $2^{-p}$ ($p \geq 0$) is defined on a standard grid. The union of same-size dyadic cubes is called a *"dyadic set"*.

$$E = \bigcup_{l=1}^{N} 2^{-p}\left(a(l) + [0,1)^n\right), \quad a(l) \in \mathbb{Z}^n$$




$p = 0$ *Side length* 1 | $p = 1$ *Side length* 1/2 | $p = 2$ *Side length* 1/4

**Theorem 13.1 (***Dyadic Cubes***):** A *"dyadic cube"* of side length $2^{-p}$ ($p \in \mathbb{N}_0$) in $\mathbb{R}^n$ is a set of the form $Q = 2^{-p}(a + [0,1)^n)$ where $a \in \mathbb{Z}^n$. We denote the set of all dyadic cubes of side length $2^{-p}$ by $\Omega_p$.

## 13.2 Inner and Outer Volumes

**Theorem 13.2 (***Inner and Outer Approximations***):** Let $A \subset \mathbb{R}^n$ be a bounded set. For $p \in \mathbb{N}_0$, we define the **inner approximation** $I(A,p)$ and the **outer approximation** $O(A,p)$ by:

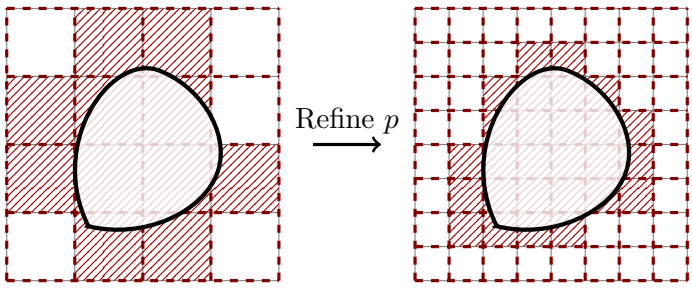
$$I(A,p) := \bigcup \{Q \in \Omega_p \mid \overline{Q} \subset A^\circ\},$$
$$O(A,p) := \bigcup \{Q \in \Omega_p \mid Q \cap A \neq \emptyset\}.$$

The **inner measure** and **outer measure** of $A$ are defined as:

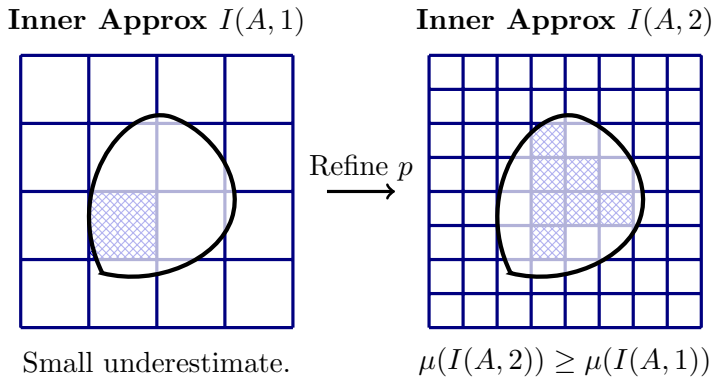
$$\mu_{\text{in}}(A) := \lim_{p\to\infty} \mu(I(A,p)), \quad \text{and} \quad \mu_{\text{out}}(A) := \lim_{p\to\infty} \mu(O(A,p)).$$

A bounded set $A$ is **Jordan measurable** if $\mu_{\text{in}}(A) = \mu_{\text{out}}(A)$.

**Outer Approx** $O(A,1)$ | **Outer Approx** $O(A,2)$



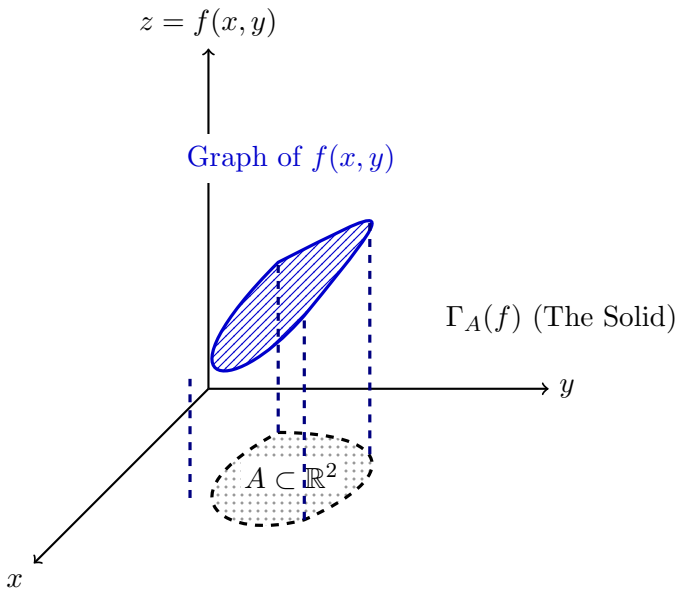

Large overestimate. | $\mu(O(A,2)) \leq \mu(O(A,1))$




# 7 Multidimensional Integrals

**Theorem 13.40** (*Hypograph*)**:** Let $A \subset \mathbb{R}^n$ and $f : A \to [0, \infty)$. The **hypograph** of $f$ is defined as the set

$$\Gamma_A(f) := \{(x, y) \in \mathbb{R}^n \times \mathbb{R} \mid x \in A, 0 \leq y < f(x)\} \subset \mathbb{R}^{n+1}.$$



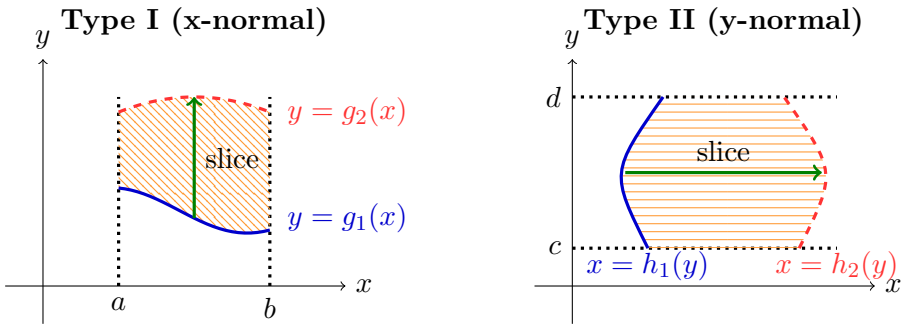

**Theorem 13.42** (*Riemann Integral as Jordan Measure*)**:** Let $A \subset \mathbb{R}^n$ be a set. A function $f : A \to [0, \infty)$ is Riemann integrable if and only if the hypograph $\Gamma_A(f)$ is Jordan measurable. In such a case, $\int_A f = \mu_{n+1}(\Gamma_A(f))$.

## 7.1 Normal Domains and Fubini's Theorem

**Theorem** (*Normal Domains / Projectable Regions*)**:** There are two primary ways to slice a 2D region $A \subset \mathbb{R}^2$:

1. **Type I (x-normal):** bounded left/right by constants $a, b$, and top/bottom by functions of $x$.

2. **Type II (y-normal):** bounded bottom/top by constants $c, d$, and left/right by functions of $y$.




**Theorem** (*Fubini's Theorem for Normal Domains*)**:** For a Type I normal domain, the double integral can be evaluated as an iterated integral:

$$\iint_A f(x,y)\,dA = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\right] dx.$$

If $A$ is Type II, integrate $dx$ first, then $dy$ on the outside. **The outermost integral bounds must always be constants!**

## 8 Change of Variables

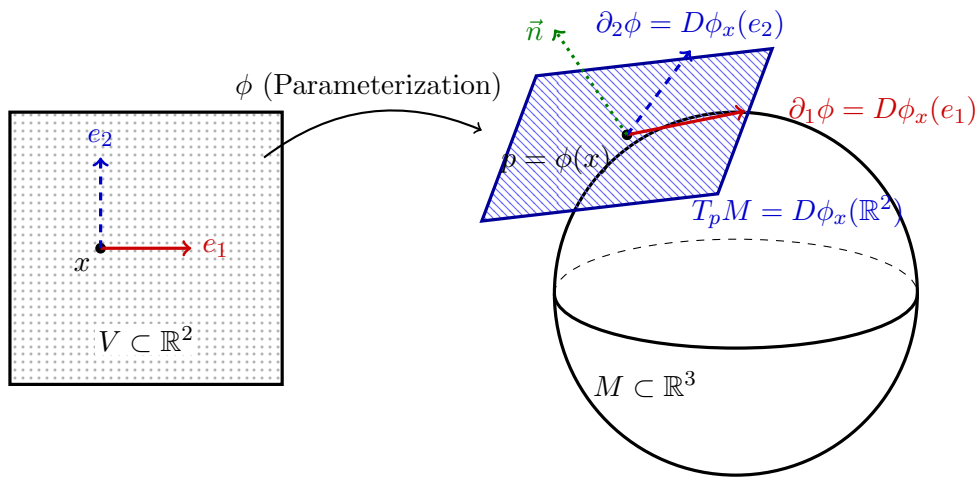
When an integral is too hard to solve in Cartesian coordinates, we distort the space using a transformation mapping $\Phi$.

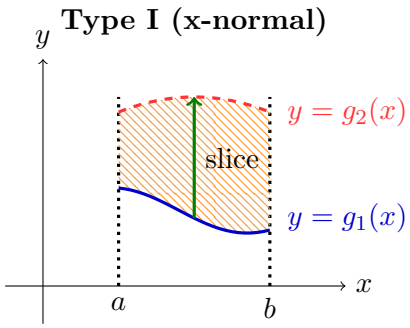
**Theorem 13.44** (*Change of Variables Formula*)**:** Suppose $U, V \subset \mathbb{R}^n$ are open sets and $\Phi : U \to V$ is a $C^1$-diffeomorphism. If $A \subset U$ is a Jordan measurable set, then

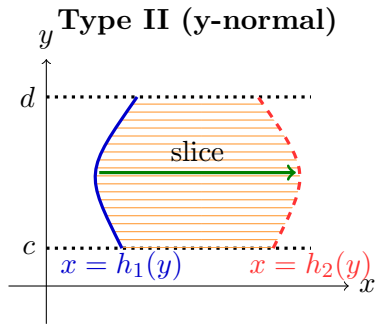
$$\int_{\Phi(A)} f(x)\,dx = \int_A f(\Phi(y))|\det J\Phi(y)|\,dy.$$

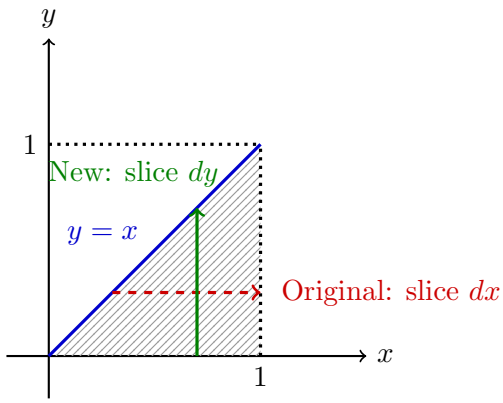
**Theorem 13.45** (*Standard Coordinates*)**:** The three standard $C^\infty$-diffeomorphisms and their Jacobians are:

- **Polar:** $\Phi(r,\theta) = (r\cos\theta, r\sin\theta)^T$. Jacobian: $r$.
- **Cylindrical:** $\Phi(r,\theta,z) = (r\cos\theta, r\sin\theta, z)^T$. Jacobian: $r$.
- **Spherical:** $\Phi(r,\theta,\varphi) = (r\sin\theta\cos\varphi, r\sin\theta\sin\varphi, r\cos\theta)^T$. Jacobian: $r^2\sin\theta$.




Type I (x-normal)
y
slice
$y = g_2(x)$
$y = g_1(x)$
x
a
b


Type II (y-normal)
y
d
slice
c
$x = h_1(y)$
$x = h_2(y)$
x


y
1
New: slice $dy$
$y = x$
Original: slice $dx$
x
1

## E-Ink Palette V2: Vibrant & Distinct

*Keine Hue-Clashes, perfekte E-Ink Lesbarkeit*

| Farbe | Code Name | Exakter Grauwert | Abstand zur Vorfarbe |
|---|---|---|---|
| | eBlack | 00.0% | – |
| | eMediumBlue | 09.2% | + 9.2% |
| | eDarkRed | 16.3% | + 7.1% |
| | eDarkGreen | 23.0% | + 6.7% |
| | eSaddleBrown | 33.0% | + 10.0% |
| | eDarkCyan | 38.2% | + 5.2% |
| | eOlive | 44.5% | + 6.3% |
| | eMediumOrchid | 50.8% | + 6.3% |
| | ePaleVioletRed | 58.0% | + 7.2% |
| | eSpringGreen | 64.4% | + 6.4% |
| | eSandyBrown | 70.7% | + 6.3% |
| | eSilver | 75.3% | + 4.6% |
| | eGold | 79.4% | + 4.1% |
| | eKhaki | 85.5% | + 6.1% |
| | eMistyRose | 91.9% | + 6.4% |